

INFORMATICS at multiple scales



Everything Computes.

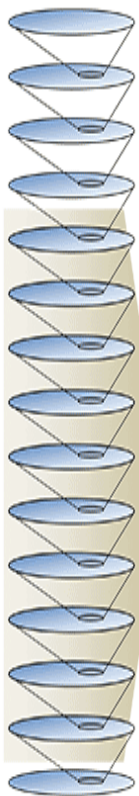


Kevin Kirby, PI (kirby@nku.edu)
James Walden, co-PI (waldenj1@nku.edu)
Maureen Doyle, co-PI (doylem3@nku.edu)



National Science Foundation
CPATH-1
CNS-0939103

The INFORMATICS AT MULTIPLE SCALES project uses the theme of scales (e.g. small–large, slow–fast, concrete–abstract, local–global) to frame a transdisciplinary vision of information and translate it into curriculum.



At the center of this effort is an experimental first-year course, **Principles of Informatics**, piloted in Fall 2010 with sixty students in the NKU College of Informatics, with majors ranging from journalism and communication studies to IT and computer science. Developing and assessing computational thinking is a key focus. In Fall 2011 it will debut in the Digitorium, a novel learning space in the new \$52M Griffin Hall informatics center.

The grant funds an **informaticist in residence** program to bring a variety of thinkers and practitioners to campus to develop transdisciplinary modules for this course and for upper level computer science classes. In 2010 the Informaticist in Residence was Anthony Moore, composer and sound theorist at the Academy of Media Arts in Cologne (*photo above*).

The grant also funds local transdisciplinary, faculty from non-informatics departments who contribute modules and teaching to the project. To date we have worked with a philosopher, an historian, and a neurobiologist.

From geospatial tagging of a renaissance Florence plague journal, to imagining Babbage and Feynman on “Rule the Air” and FCC spectrum allocation, these visitors are bringing engaging new ways of thinking about computing to students in a metropolitan university.

These modules are growing into a book and a set of associated online resources, to appear in 2012.

Kirby, K. Walden, J., Garns, R., Doyle, M.,
“**The Great Chains of Computing:**
Informatics at Multiple Scales.”
Proc. Foundations of Information Science,
Beijing, August 2010.



scalinginformatics.nku.edu